

Algebra 1
Unit 9
Lesson 2 Practice

Name Answer key
Date _____
Period _____

Use the information provided in each graph to write the equations in vertex form and standard form.

1. Identify the points.

a. Vertex: (0, -2)

b. Point: (2, 6)

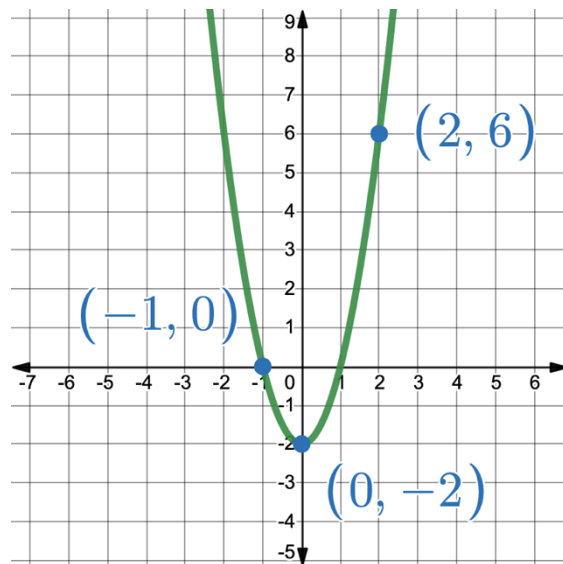
2. Write the equation for this quadratic in vertex form and standard form.

- a. Vertex form:

$$f(x) = 2(x - 0)^2 - 2$$

- b. Standard form:

$$f(x) = 2x^2 - 2$$



3. Identify the points.

a. Vertex: (2, -2)

b. Point: (4, -1)

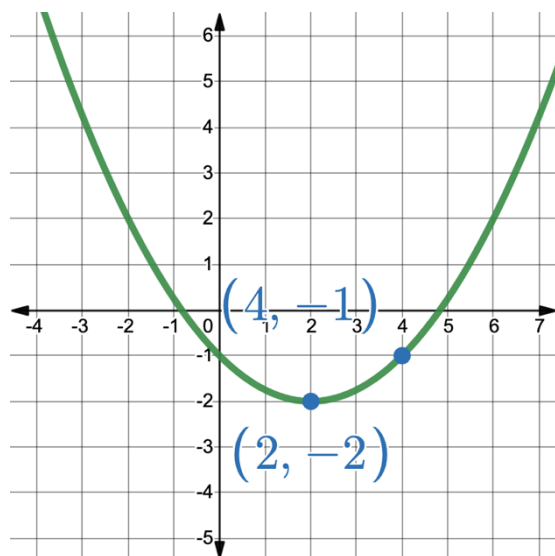
4. Write the equation for this quadratic in vertex form and standard form.

- a. Vertex form:

$$f(x) = \frac{1}{4}(x - 2)^2 - 2$$

- b. Standard form:

$$f(x) = \frac{1}{4}x^2 - x - 1$$



5. Identify the points.

a. Vertex: (0, -6)

b. Point: (-1, -2)

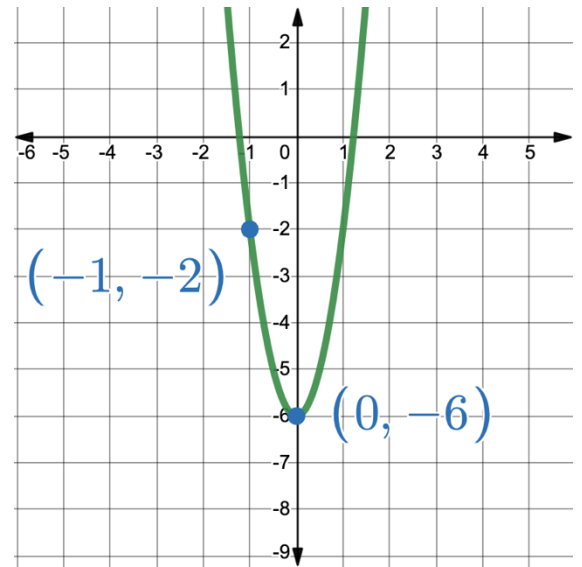
6. Write the equation for this quadratic.

a. Vertex form:

$$f(x) = 4(x - 0)^2 - 6$$

b. Standard form:

$$f(x) = 4x^2 - 6$$



7. Identify the points.

a. Vertex: (-3, 0)

b. Point: (-6, -4.5)

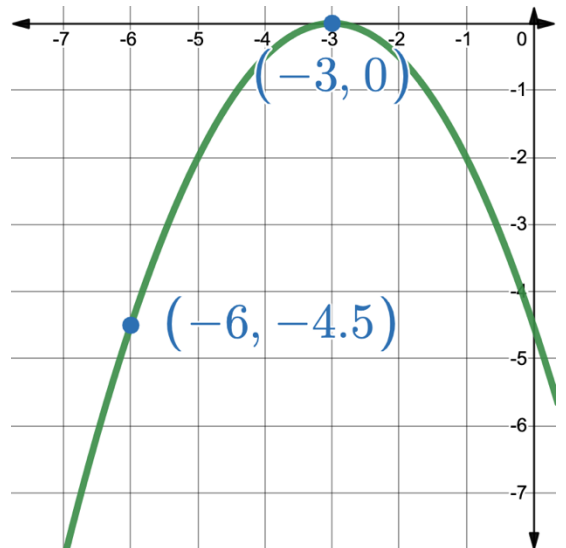
8. Write the equation for this quadratic.

a. Vertex form:

$$f(x) = -\frac{1}{2}(x + 3)^2$$

b. Standard form:

$$f(x) = -\frac{1}{2}x^2 - 3x - 4.5$$



9. Identify the points.

a. Vertex: (3, -1)

b. Point: (4, 1)

10. Write the equation for this quadratic.

a. Vertex form:

$$f(x) = 2(x - 3)^2 - 1$$

b. Standard form:

$$f(x) = 2x^2 - 12x + 17$$

